Smooth Path Planning and Dynamic Contact Force Regulation for Robotic Ultrasound Scanning

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***Abstract*—The robotic breast ultrasound scanning (RBUS) system can help sonographer in the early screening of breast cancer. However, it still faces rigorous safety concerns, including non-smooth scanning paths and improper contact force regulation. In this letter, we propose a new path optimization method to improve the scanning smoothness, as well as, a dynamic contact regulation strategy considering both the breast deformation and ultrasound images. For the path planning, the anti-radial scanning approach is first adopted to generate some initial path points together with their normal vectors; then, a dual-objective optimization function is formulated, which considers both the acceleration continuity and deviation of the path, to filter out local path anomalies. For the force control, a contact force-strain regression model is first proposed and used to predict the desired force between the breast and probe at each path point. While within the adjacent path points, an updating algorithm is introduced to dynamically adjust the contact force in real-time by surveying the confidence of the ultrasound image. Our experimental results show that the proposed methods can make the scanning path smoother, and obtain intimate probe-tissue contact with less desired force.**

***Index Terms*—Robotic breast ultrasound scanning, path optimization, variable desired force control.**

I. Introduction

BREAST cancer is a life-threatening disease, with over 2 million new cases and more than 684,000 deaths reported annually worldwide [1]. Advances in science and technology have led to significant improvements in breast cancer screening methods. Among these, breast ultrasound (BUS) has gained prominence due to its non-invasive nature and diagnostic efficiency. However, the effectiveness of BUS heavily depends on the sonographer's expertise [2], [3]. Furthermore, the examination process is both time-consuming and physically strenuous, contributing to a reported 90% prevalence of work-related musculoskeletal disorders among sonographers [4], [5]. To address these limitations, robotic breast ultrasound scanning (RBUS) technology has been developed. This innovation standardizes and simplifies the examination process, increasing the efficiency of ultrasound examinations [6].

Automatic path planning is a critical component of RBUS technology, playing a central role in optimizing ultrasound examinations. Path planning methods can be categorized into manual, semi-automatic, and fully automatic approaches [6], [7], [8], [9]. Recent advancements have focused on the automatic selection of region of interest (ROI) and the generation of efficient scanning paths [10]. These systems often utilize RGB-D cameras to capture 3D point cloud data from the human body surface, enabling automated ROI segmentation and extraction [11], [12], [13], [14], [15]. For instance, Wang et al. [11] and Huang et al. [13] directly applied point cloud data to generate robotic path, streamlining the processing workflow. However, this approach may introduce path fluctuations due to irregularities in the point cloud data, potentially causing larger accelerations that lead to patient discomfort. To address this issue, Tan et al. [15] employed a non-uniform rational B-spline (NURBS) curve fitting to refine the point cloud data, reducing path irregularities and ensuring smoother robotic motion. For effective path planning, smooth paths are essential to minimize abrupt accelerations and decelerations, ensuring stable and efficient robot movement.

In automatic RBUS technology, maintaining stable contact between the ultrasound probe and tissue is a critical challenge [11]. To address this, Wang et al. [11] developed an admittance control algorithm based on a force-velocity hybrid control framework, enabling stable and consistent force control in an automatic RBUS system. Similarly, Tan et al. [15] introduced a fully automated RBUS system that incorporates PID control, a dual robotic arm configuration, and a flexible ultrasound probe clamping device. These systems not only ensures stable and constant contact force but also improves scanning efficiency. In conventional BUS, sonographers must dynamically regulate the probe's contact force based on real-time image feedback. While minimizing contact force is essential for ensuring patient safety, it is equally important to maintain an adequate level of contact force to achieve high-quality ultrasound images. The desired force may vary depending on the specific application, which refers to the target contact force value required to achieve safe and intimate probe-tissue contact and is typically greater than 0 N. Welleweerd et al. [16] performs visual servoing control based on real-time ultrasound confidence maps to optimize probe-breast contact and reduce tissue

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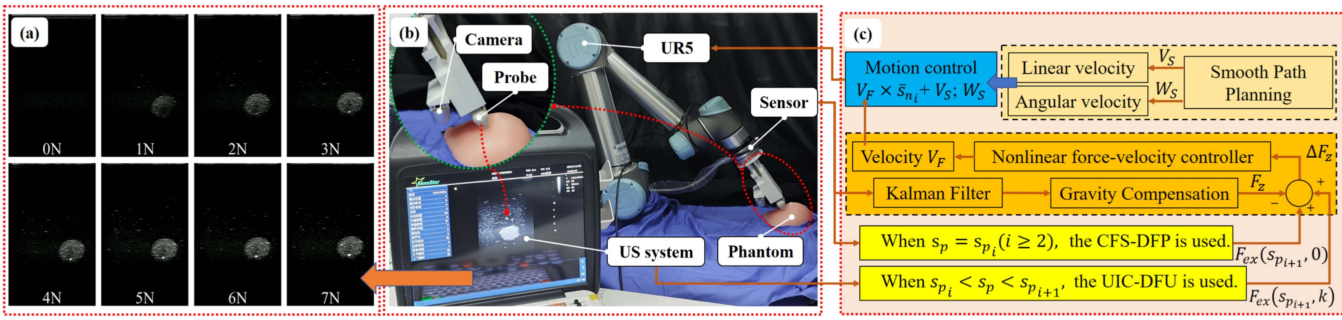

Fig. 1. (a) shows ultrasound images at the same location with different contact forces, (b) shows the RBUS system, and (c) shows control flowchart of automatic RBUS.

deformation. However, this system relies entirely on ultrasound image confidence (rather than force sensors) to control contact, which cannot achieve stable contact force control and may pose safety hazards. Despite the critical importance of this adjustment process, existing research on automatic RBUS has not extensively explored real-time dynamic control of contact force. However, similar studies have been conducted in other fields of ultrasound scanning. For instance, Tsumura et al. [17] systematically evaluated how varying contact forces impact the quality of thyroid ultrasound image and patient comfort, providing valuable experimental insights. However, their study did not address real-time dynamic force adjustments. In contrast, Zakeri et al. [18] proposed an intelligent interaction force control method for a cardiac ultrasound robotic system. This system uses artificial intelligence to dynamically adjust the contact force based on real-time ultrasound image feedback, thereby enhancing image quality. The hearts of different patients have similar structural features, but the internal structures of the breasts of different patients vary greatly, so this method cannot be transferred to breast ultrasound examinations.

This study aims to achieve smooth path planning and dynamic contact regulation, with the goal of improving the safety and obtaining intimate probe-tissue contact with less desired force. The main contributions of this letter are as follows: (i) a dual-objective path optimization method (DO-ACDC) is proposed, considering path acceleration continuity and deviation control. This method aims to optimize the smoothness of the initial scanning path and enhance angular acceleration stability; (ii) a contact force-strain regression model-based desired force prediction algorithm (CFS-DFP) is introduced to dynamically adjust the robot's desired force at each path point; (iii) a ultrasound image confidence-based desired force update algorithm (UIC-DFU) is proposed to adjust the desired force between adjacent path points.

II. Ultrasound Scanning Path Planning and Optimization

In clinical practice, various breast ultrasound scanning approaches are employed, with the anti-radial scanning approach recognized as one of the most effective methods [19]. This technique involves scanning from the periphery of the breast toward its center, which helps minimize tissue folding and

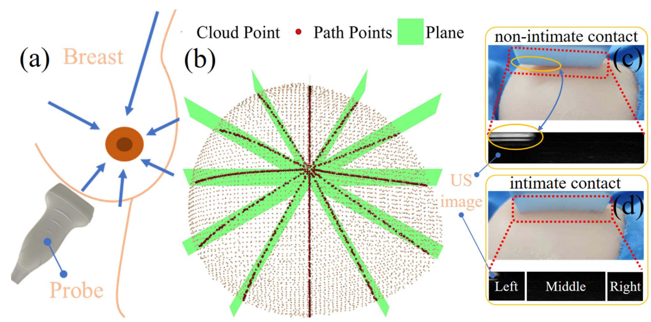

Fig. 2. (a) shows schematic diagram of ultrasound scanning method, (b) shows point cloud model, (c) and (d) show probe-tissue contact and corresponding top of US image.

maximizes the tension of Cooper's ligaments, thereby enhancing the efficiency of the examination, as illustrated in Fig. 2(a). Furthermore, any positive findings must be evaluated within the anti-radial scanning to ensure accurate assessment.

A. Path Planning

As shown in Fig. 1(b), the breast phantom is placed on a tabletop and covered with a medical sterile cloth, exposing only the region to be scanned. Following the method proposed by Wang et al. [11], multi-angle point clouds are captured using a RealSense D435i depth camera mounted on a UR5 robotic arm operated in zero-force drag mode, and subsequently segmented based on color features to extract only the phantom region. A complete point cloud model of the phantom is then registered through the Point-to-Plane ICP algorithm, achieving a registration accuracy of RMSE $= 0.36$ mm [20], as illustrated in Fig. 2(b). In this study, the phantom point cloud model is sliced to extract cross-sectional points [21]. Starting from the highest point (nipple position) of the point cloud, planes are systematically arranged at angular intervals of θ_{us} in a counterclockwise direction, as shown in (1).

$$\theta_{us} = 2 \cdot \arcsin\left(\frac{L}{2r}\right) \tag{1}$$

where r represents the maximum radius from the highest point in the point cloud to the edge, L denotes the imaging width of the probe, and then the number of anti-radial scanning paths $n_{us} = 2\pi/\theta_{us}$. The intersection points between these planes and the point cloud are then identified and selected as the initial path points, as depicted in Fig. 2(b).

The path P is generated using the plane slicing method described above, where the normal of each path point are determined by calculating the fitted plane normal formed by its neighboring point clouds [11]. However, the raw point cloud data obtained from the depth camera often exhibits fluctuations. Using this unprocessed data directly for robotic ultrasound scanning can result in velocity instability. To address this issue, NURBS curve fitting is applied to smooth the path, reduce data complexity, and improve path controllability and noise resistance. This process effectively minimizes the variations and fluctuations in the initial path points, ensuring a more stable and efficient scanning path [12].

The recursive definition of the NURBS basis function $N_{i,k_p}(u)$ is given by:

$$N_{i,k_p}(u) = \frac{u - u_i}{u_{i+k_p} - u_i} N_{i,k_{p-1}}(u) + \frac{u_{i+k_{p+1}} - u}{u_{i+k_{p+1}} - u_{i+1}} N_{i+1,k_{p-1}}(u) \tag{2}$$

where k_p is the degree of the NURBS curve. When $k_p = 3$, it ensures the second-order continuity of the scanning path. The discrete parameter values u_i correspond to the discrete points on the curve, i.e., the initial path point, while the curve parameter u is used to describe continuous positions along the curve.

For a set $U = \{u_{p_1}, u_{p_2}, \ldots, u_{p_q}; u_{n_1}, u_{n_2}, \ldots, u_{n_q}\}$, where u_{p_i} and u_{n_i} represent the path point and normal vector, respectively. The goal of fitting the NURBS curve is to minimize the deviation between the path points and the curve by optimizing the control point set.

$$\min \sum_{i=1}^{n} \|P_i - C(u_i)\|^2 \tag{3}$$

The definition of the NURBS curve is given as:

$$C(u) = \frac{\sum_{j=0}^{m} w_j N_{j,k_p}(u) e_j}{\sum_{j=0}^{m} w_j N_{j,k_p}(u)} \tag{4}$$

where w_j represents the weight of the control point e_j, m is the number of control points, and $N_{j,k_p}(u)$ is the j-th basis function of degree k_p.

By constructing the basis function matrix $N_{ij} = N_{j,k_p}(u_i)$, the minimization problem of the objective function can be reformulated as solving the linear system $N \cdot E = P$. Subsequently, the positions of the control points are optimized using the least-squares method to ensure the fitting accuracy of both the path points and the normal vectors.

NURBS curve fitting can effectively suppress "deviations and fluctuations" caused by irregular initial path points to a certain extent. However, certain localized regions of the curve may still exhibit such issues. Therefore, the generated path requires further smoothing. In the next section, a smoothing optimization method was proposed to address this problem.

B. Path Optimization

1) Generation of Regular Surface Based on Path Point and Normal Vector: Due to potential inconsistencies in the parameter ranges of the path points and normal vectors, it is necessary to first synchronize the parameters to establish a coupling relationship. This is expressed as:

$$w_i = \max(u_{\min}) + \frac{i-1}{m_w - 1}\Delta u, \quad i = 1, \ldots, m_w \tag{5}$$

Where $\Delta u = \min(u_{\max}) - \max(u_{\min})$, $u_{\min} = \min(u_p, u_n)$, $u_{\max} = \max(u_p, u_n)$, and m_w is set as the number of initial path points. Next, a regular surface can be constructed based on the path point curve $C_p(w)$ and the normal vector curve $C_n(w)$, which is represented as:

$$s(w, v) = (1 - v)C_p(w) + vC_n(w) \tag{6}$$

where $C_p(w)$ and $C_n(w)$ represent the corresponding path point and normal vector on the fitted curve from (4), v is the blending parameter, and $s(w, v)$ is the resulting regular surface.

2) Establishment of a Dual-Objective Function for Surface Optimization: To ensure the smoothness of the surface, acceleration continuity is used as the metric. The acceleration continuity is computed through the square integral of the second-order derivatives of the surface $s(w, v)$, expressed as:

$$E_{acc} = \iint_D \left(\left\| \frac{\partial^2 s(w,v)}{\partial w^2} \right\|^2 + 2 \left\| \frac{\partial^2 s(w,v)}{\partial w \partial v} \right\|^2 + \left\| \frac{\partial^2 s(w,v)}{\partial v^2} \right\|^2 \right) dw\, dv \tag{7}$$

In addition to ensuring surface smoothness, it is essential to ensure that the fitted surface does not deviate excessively from the initial path. Therefore, a deviation control term is introduced, defined as:

$$E_{\text{error}} = \alpha \sum_{i=1}^{q} \|u_{p_i} - C_p(w_i)\|^2 + (1-\alpha) \sum_{i=1}^{q} \|u_{n_i} - C_n(w_i)\|^2 \tag{8}$$

Where α and $1 - \alpha$ are the weights.

By considering acceleration continuity and deviation control, the dual-objective function is constructed to balance the surface smoothness and fitting accuracy:

$$\min \quad E_{all} = \delta E_{acc} + (1 - \delta) E_{error} \tag{9}$$

where δ represents the weight for acceleration continuity, controlling the surface smoothness, and $1 - \delta$ represents the weight for deviation control, managing the fitting accuracy.

An improved quasi-Newton optimization algorithm, specifically the L-BFGS-B method, is employed to minimize the objective function (9). Finally, the optimized set of path is obtained as $S = \{s_{p_1}, s_{p_2}, \ldots, s_{p_q}; s_{n_1}, s_{n_2}, \ldots, s_{n_q}\}$.

III. Variable Desired Force Control

As illustrated in Fig. 1(a), the applied contact force at a given scanning position clearly influences ultrasound image quality.

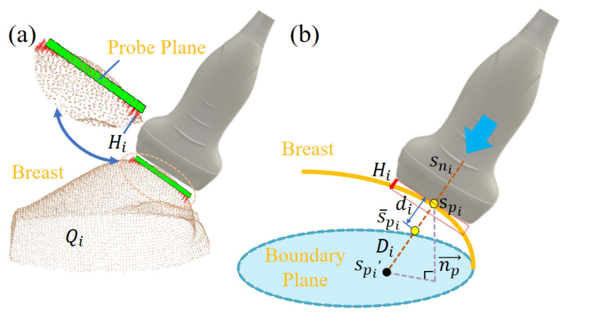

Fig. 3. (a) shows schematic diagram of projection depth H_i,(b) shows schematic diagram of probe-tissue interaction.

While increasing contact force enhances image clarity and improves tumor boundary delineation, excessive force beyond 5 N fails to yield additional image quality improvements. Moreover, such excessive force may induce substantial breast tissue deformation and cause patient discomfort. Our experimental results and analysis demonstrate the critical importance of precise force regulation. Therefore, the subsequent sections will focus on investigating real-time dynamic control strategies for contact force in automated robotic breast ultrasound (RBUS) systems.

A. Contact Force-Strain Regression Model-Based Desired Force Prediction Algorithm

1) Mathematical Model Based on Deformation Estimation: Researchers often consider the boundary of the breast as a plane [11], [22], [23]. In this work, the boundary plane of the breast phantom is obtained through the following steps: first, the entire point cloud of the phantom is projected onto the tabletop to extract its two-dimensional boundary curve; subsequently, this 2D boundary is mapped back to three-dimensional space using nearest-neighbor search to obtain the point cloud representing the bottom edge of the phantom; finally, the boundary plane is fitted to these edge points. As shown in Fig. 3(b), s_{p_i} and s_{n_i} represent the path point and normal vector of the path point, $\overrightarrow{n_p}$ represents the normal vector of the plane, and $s_{p_i}{}'$ represents the projection of the path point s_{p_i} onto the boundary plane along the direction of the normal vector s_{n_i}. $\bar{s}_{p_i}$ represents the point corresponding to the probe compression depth d. D can be expressed as the distance from the path point to the boundary plane along the normal vector.

The mechanical properties of the breast tissue are hyperelastic [23]. Considering the tissue's heterogeneity, machine palpation is performed at three different thicknesses (D) of the breast. Let d represent the probe compression depth, then the strain can be expressed as $\varepsilon = d/D$. Assuming that the relationship between strain and the corresponding contact force follows a cubic polynomial:

$$F_c = \beta_3\varepsilon^3 + \beta_2\varepsilon^2 + \beta_1\varepsilon + \beta_0 \tag{10}$$

where β_0, β_1, β_2, and β_3 are elastic parameters, and F_c is the predicted contact force.

Given the complex surface characteristics of the breast and the varying compression depths d and projection thicknesses D required at different locations of the probe, a local deformation measure is defined as $\gamma = d/H$. Where d represents the probe compression depth under a given force, and H represents the projection depth which is the projection depth of the ultrasound probe's bottom surface along the normal vector s_{n_i}, such that the ultrasound probe's bottom surface just fully contacts the breast surface, as shown by the red line segment H_i in Fig. 3(a). To compute the projection depth for each path point, a plane is established at the probe's bottom using the s_{p_i} and s_{n_i}. Each point Q_i in the point cloud is projected onto this plane. As shown in Fig. 3(a), the green rectangle is used to define the boundary of the probe's bottom. By determining whether the projection coordinates of point Q_i lie within the rectangle, the distance between Q_i and its projected point is calculated. Finally, the maximum distance is taken as the projection depth H_i. Based on the above calculations, the projection depths H_i and projection thicknesses D_i are known for all path points on the every path.

2) Desired Force Prediction Algorithm At Each Path Point: The female breast exhibits considerable variability in size, shape, and elastic properties among individuals, the desired force may vary for different breasts. At the first path point s_{p_1} of the first path, the probe automatically compresses the phantom along the normal direction s_{n_1} until a complete and clear ultrasound image is obtained. At this moment, the force/torque sensor is automatically recorded and the contact force is logged as the initial desired force for every path. When $s_p = s_{p_i}$ ($i \geq 2$, the path point of every path), the contact force F_z is computed based on the current force/torque sensor reading, as shown in Fig. 1(b). The strain ε_i at this point is then determined by solving the contact force-strain regression model from (10). It should be noted that cubic polynomials may have multiple solutions, but within the research background of this article, the actual reasonable solution is greater than zero and is the smallest solution. Then the probe compression depth $d_i = \varepsilon_i \cdot D_i$ and the local deformation $\gamma_i = d_i/H_i$ are calculated. Given the continuity of deformation, it is assumed that the local deformation at two consecutive path points is equal, i.e., $\gamma_{i+1} = \gamma_i$. This assumption enables the prediction of the required compression depth $d_{i+1} = \gamma_{i+1} \cdot H_{i+1}$ and strain $\varepsilon_{i+1} = d_{i+1}/D_{i+1}$ at next path point. Then the desired force $F_{ex}(s_{p_{i+1}}, k)$ (where $k = 0$) at the next path point $s_{p_{i+1}}$ is calculated from (10).

B. Ultrasound Image Confidence-Based Desired Force Update Algorithm

1) Confidence-Based Probe-Tissue Contact Assessment Model: When the ultrasound probe is not in contact with the breast, a white strip appears at the top of the ultrasound image. When the probe is in intimate contact with the breast, this white strip disappears, and the image becomes complete [16], as shown in Fig. 2(c) and (d). In order to quantify the contact between the probe and the tissue, this letter introduces confidence maps. Confidence maps provide a pixel-wise estimation of uncertainty in ultrasound (US) images. Given an image $I : \Omega \rightarrow [0, 1]$, a corresponding confidence map $C : \Omega \rightarrow [0, 1]$ assigns to each pixel $(i, j) \in \Omega := [1..n_{us}] \times [1..m_{us}]$ a confidence value indicating the reliability of $I(i, j)$. The confidence map C is computed based on the probability of a random walk originating from a pixel (i, j) and reaching virtual transducer elements [24], thereby reflecting the probe-tissue contact quality [25], [26], [27]. When contact is intimate, confidence values are high(approaching 1).

Based on the characteristics of the ultrasound image, the upper region of the image is divided into three parts: left, middle, and right, as shown in Fig. 2(d). Confidence values for each part and the overall parts are obtained using a confidence algorithm, represented as CL_{mean}, CM_{mean}, CR_{mean}, and $C_{\text{mean\_part}}$, where $C_{\text{mean\_part}} = (CL_{\text{mean}} + CM_{\text{mean}} + CR_{\text{mean}}) / 3$. When these three values reach a desired value (related to the material of breast phantom), it indicates that the left, middle, and right parts of the probe are all in intimate contact with the breast, resulting in a complete ultrasound image [27].

2) Desired Force Update Algorithm Between Path Points: When the US probe scans between adjacent path points, the appropriate desired force required is not constant. In this letter, real-time feedback from ultrasound images is used to adjust the desired force between two path points online. When $s_{p_i} < s_p < s_{p_{i+1}}$, i.e., between two adjacent path points, the UIC-DFU proposed in this letter can be expressed as:

$$F_{ex}(s_{p_{i+1}}, k{+}1) = \left(\frac{(K \cdot \Delta C_{mean})^3}{1{+}(K \cdot \Delta C_{mean})^2} {+}1 \right) \cdot F_{ex}(s_{p_{i+1}}, k) \tag{11}$$

where $F_{ex}(s_{p_{i+1}}, k)$ represents the desired force at the k-th time step between path points s_{p_i} and $s_{p_{i+1}}$, and K is a scaling parameter that reflects the speed of adjustment in desired force. All other conditions remain unchanged, the larger K, the greater the change in desired force, resulting in the actual confidence approaching the desired confidence faster. The term ΔC_{mean} represents the difference between the actual confidence and the desired confidence, defined as:

$$\Delta C_{\text{mean}} = \max\left(0, \min\left(CL_{\text{mean}}, CM_{\text{mean}}, CR_{\text{mean}}\right) - C_{ex}\right) \tag{12}$$

where C_{ex} represents the desired confidence, which indicates a higher quality of contact. Analysis of (11) and (12) reveals that the desired force varies gradually when actual confidence approaches desired confidence, but changes more rapidly when they diverge clearly. This online force adaptation mechanism leverages ultrasound image confidence to dynamically optimize the desired force for obtaining intimate probe-tissue contact.

C. RBUS System Workflow

As shown in Fig. 1(b) and (c), the robot, carrying the ultrasound probe, performs the ultrasound scanning along the optimized path obtained in Section II. The force/torque sensor records force information in real-time, and through kalman filtering and gravity compensation, the true contact force F_z is obtained. During the scan between adjacent path points ($s_{p_i} < s_p < s_{p_{i+1}}$), the desired force $F_{ex}(s_{p_{i+1}}, k)$ at time k is updated online using the UIC-DFU. When the scan reaches the path point $s_{p_{i+1}}$, the desired force $F_{ex}(s_{p_{i+2}}, 0)$ for the next path point $s_{p_{i+2}}$ is updated using the CFS-DFP. Therefore, the difference between the desired force and the true contact force is represented as:

$$\Delta F_z = \begin{cases} F_{ex}(s_{p_{i+1}}, k) - F_z, s_{p_i} \le s_p \le s_{p_{i+1}} \text{ and } k \ge 1 \\ F_{ex}(s_{p_{i+2}}, 0) - F_z, s_p = s_{p_{i+1}} \end{cases} \tag{13}$$

Then, ΔF_z is converted into the probe's normal velocity V_F through nonlinear force-velocity admittance control, expressed as:

$$V_F = M \times \left(\frac{2}{1 + e^{-\rho \cdot \Delta F_z}} - 1 \right) \tag{14}$$

where M is the maximum linear velocity, and ρ is the contact force change rate.

By obtaining the probe's pose, its normal vector $\bar{s}_n$ can be calculated, and the robot's linear velocity V and angular velocity W can be expressed as:

$$\begin{bmatrix} V \\ W \end{bmatrix} = \begin{bmatrix} V_F \times \bar{s}_{n_i} + V_S \\ W_S \end{bmatrix} \tag{15}$$

where V_S and W_S represent the linear and angular velocities, and are calculated based on the optimized path [11].

Through the aforementioned nonlinear force-velocity admittance control (low-level controller), stable contact force control can be achieved. Meanwhile, using the CFS-DFP and UIC-DFU (high-level controllers), the desired force is updated online, achieving variable desired force contact force control, which balances probe-tissue contact quality and patient comfort.

IV. Experiments and Results Analysis

A. System Setup

The proposed methods and algorithms in this study were validated on the automatic RBUS system, as illustrated in Fig. 1(b). This system consists of five primary components: a robotic arm (UR5, Universal Robots, Denmark), an ultrasound imaging system equipped with a linear probe (SonoStar C10 Wired Probe, SonoStar, China), a force/torque (F/T) sensor (KWR75 A, Kunwei Technology, China), an RGB-D camera (Realsense D435i, Intel, USA). Robot path planning and control were implemented in Python on a laptop powered by an Intel Core i9-13900HX 2.2 GHz processor with 16 GB RAM. The ultrasound probe was mounted at the end-effector of the robotic arm using a 3D-printed fixture. The F/T sensor was installed between the robot's end-effector and the probe fixture to measure the contact force between the ultrasound probe and the phantom. Additionally, the RGB-D camera was attached to the robot's end-effector and positioned adjacent to the probe for capturing the point cloud data of the breast phantom, as shown in Fig. 1(b). The robotic arm operated at a control frequency of 125 Hz, while the force sensor sampled contact force signals at a frequency of 1 kHz.

B. Verification of Path Planning and Optimization

As shown in Fig. 2(b), cross-sectional points were obtained by slicing the complete point cloud model of the phantom with a plane($\theta_{us} = 30^\circ$ in (1)) and the normal vectors are estimated

TABLE I
Overall Mean and Standard Deviation of Path Deviation and Angle Increment

Method	Deviation (10^{-2} mm)	Angle Increment ($10^{-1\circ}$)
Initial	33.0±23.0	7.49±9.63
NFO	3.7±2.7	0.31±0.51
EBTSO	3.1±2.4	0.17±0.30
DO-ACDC	2.9±2.1	0.10±0.12

according to Section II-A. In this experiment, the same initial path were optimized using three methods including NURBS-Fitting Optimization [28] (NFO, the degree $k_p = 3$, the number of control points $m = 8$, and the weights of the control points $w_j = 1$ in (4)), an energy-based path smoothing optimization (EBTSO) proposed by Wang et al. [29], and DO-ACDC(the weight $\alpha = 0.5$ and $\delta = 0.5$ in (8) and (9).

To quantitatively evaluate the optimization effects of the three methods on path points, the distance deviation $dist_i$ is defined as the vertical distance between the path point and the plane slicing, where the positive and negative signs of $dist_i$ represent the path points on both sides of the plane, respectively. As shown in Table I, compared with NFO, DO-ACDC reduces the mean $|dist_i|_{mean}$ by 21.3% and the mean $|dist_i|_{std}$ by 22.3% for all paths. It also achieves a 6.7% lower mean $|dist_i|_{mean}$ and 13.8% lower mean $|dist_i|_{std}$ compared to EBTSO. These results show that the DO-ACDC-optimized path better approximates the ideal path.

To quantitatively evaluate the optimization effects of the three methods on the normal vectors, the angle increment $\Delta\theta_i$ was used as an evaluation metric for path smoothness:

$$\Delta\theta_i = \theta_{i+1} - \theta_i = \arccos(\mathbf{s}_{n_{i+1}} \cdot \mathbf{s}_{n_i}) - \arccos(\mathbf{s}_{n_i} \cdot \mathbf{s}_{n_{i-1}}) \tag{16}$$

where $\mathbf{s}_{n_i}$ represents the normal vector at the i-th point. As shown in Table I, compared with NFO, DO-ACDC reduces the mean $|\Delta\theta_i|_{mean}$ by 67.7% and the mean $|\Delta\theta_i|_{std}$ by 41.2% for all paths. It also achieves a 76.5% lower mean $|\Delta\theta_i|_{mean}$ and 60.0% lower mean $|\Delta\theta_i|_{std}$ compared to EBTSO. These results demonstrate that the proposed DO-ACDC has a greater advantage in improving the smoothness of the initial paths. It should be noted that the dense distribution of path points inherently limits the magnitude of angular increments, making even small numerical reductions meaningful in practice.

To intuitively compare the optimization performance of different methods on the initial path points, this study visualizes the optimization results of Path 1 and Path 10. Path 1 exhibits larger curvature variations, whereas Path 10 shows relatively smoother curvature. As shown in Fig. 4, after the optimization method proposed in this letter, the initial path has clearly reduced the clutter and mutation of path points and normals, and the distribution is more reasonable, uniform and smooth. In terms of angular increments, DO-ACDC yields smaller angular increments than NFO and EBTSO, indicating superior optimization performance. For the last 20 points of the paths near the nipple, the $\Delta\theta_i$ optimized by DO-ACDC were clearly lower than those optimized by NFO and EBTSO. Specifically, for the last 20

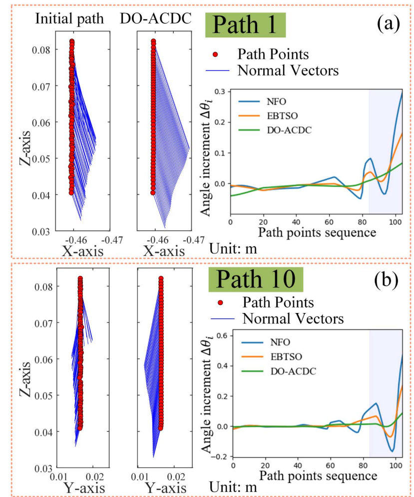

Fig. 4. Experimental results of path optimization of two paths.

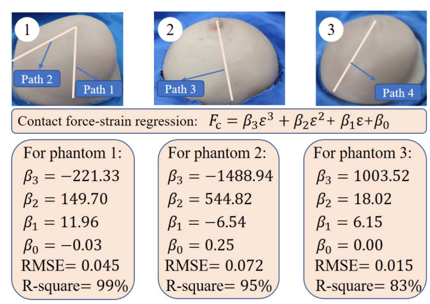

Fig. 5. Three different phantoms and their contact force-strain regression models.

points of Path 1, the $|\Delta\theta_i|_{\text{mean}}$ were 0.086°, 0.052°, and 0.032°, respectively, while for Path 2, the $|\Delta\theta_i|_{\text{mean}}$ were 0.129°, 0.065°, and 0.018°. This smoothing reduces abrupt path changes for better scanning stability, with acceptable loss of fine detail near the nipple region given the contact compression depth.

C. Validation of Variable Desired Force Control

The female breast shows vast variations in size, shape, and elastic properties from patient to patient. As shown in Fig. 5, three breast phantoms are used to validate the variable desired force control. The first phantom (LYD810, Liuyedao Medical Technology, China) is made of silicone. The second phantom (YB-RX311, Shenzhen Yichuang Medical Technology Co., China) is also made of silicone but is softer than the first and has a hemispherical shape, which differs clearly from the first. The third phantom (SE-IG0006-S1, Zhejiang Yizai Medical Technology Co., China) has a hydrogel core covered with a silicone skin, making it the softest among the three, with a shape similar to the first. Considering the heterogeneity of tissue properties, the probe was compressed at a speed of 1 mm/s at three different positions of the each phantom, while the probe's compression depth and the corresponding contact force were recorded. The fitting results of strain and the corresponding contact force are shown in Fig. 5, indicating a good fit. Using the methods for

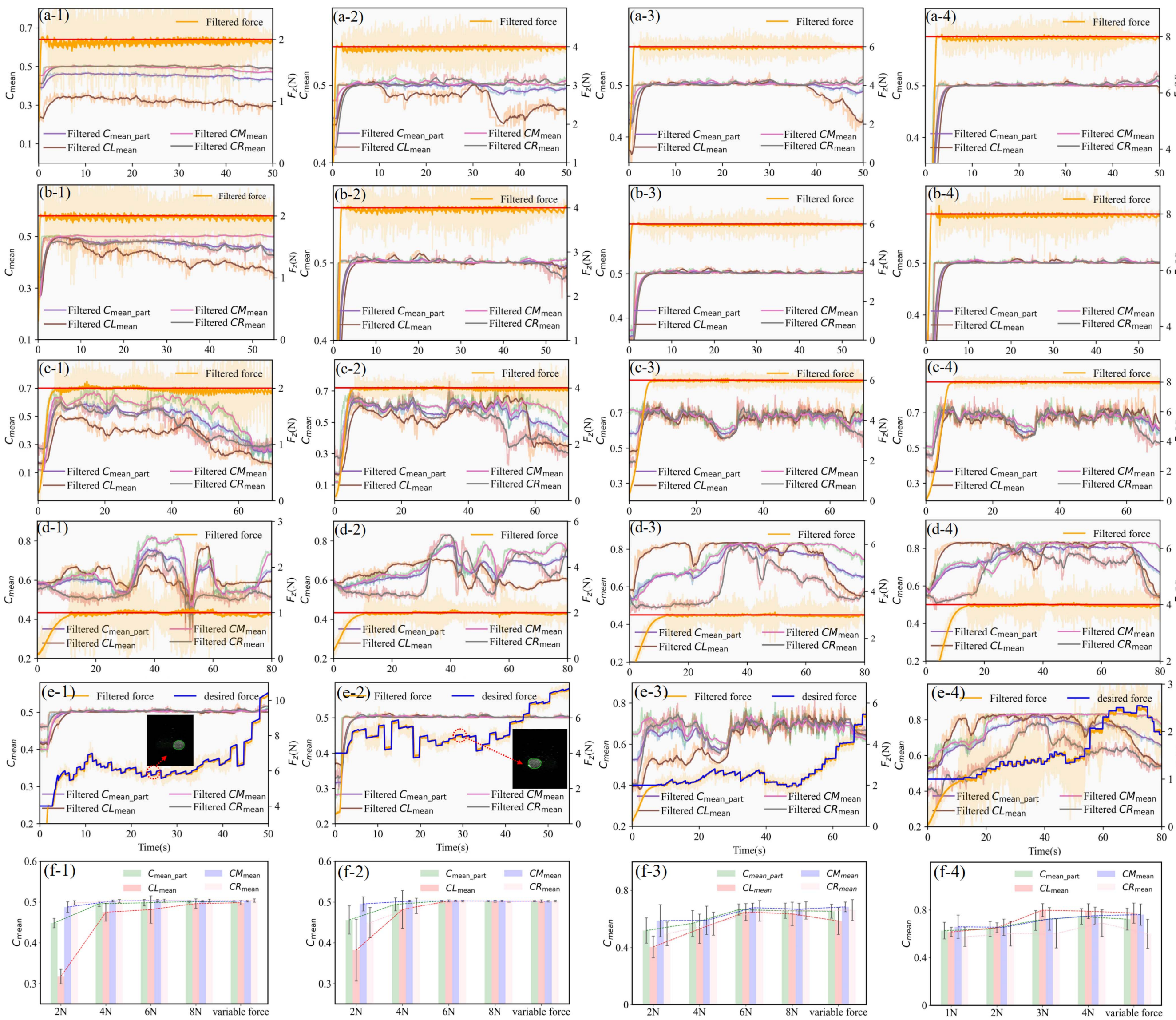

Fig. 6. Experimental results of variable desired force control and constant desired force control. (a), (b), (c) and (d) show the confidence curves and contact force curves for different constant desired forces control on the four paths, respectively. (e) shows the confidence curves and contact force curves for variable desired force control on the four paths, respectively. In (e-1) and (e-2), the ultrasound images with lesions obtained during the scanning process, along with the associated contact force areas, are presented. The lesion contours, obtained via a lesion segmentation algorithm, are visualized. (f) displays the bar plots with error bars of confidence values for the four different paths, respectively.

calculating the projection depth H and projection thickness D described in Section III-A, the values of H_i and D_i along the four paths optimized by DO-ACDC(from Section II-B) were calculated.

To compare the differences between constant desired force control and the proposed variable desired force control during automatic RBUS both constant force control and variable desired force control were applied along the four paths optimized by DO-ACDC. The scanning speed was set to $V_S = 1$ mm/s, the force control parameters $M = 0.005$, $\rho = 3$ in (14), and the desired confidence $C_{ex} = 0.52$ in (12), with a confidence variation rate $K = 6$ in (11). As shown in Fig. 6(a), (b), (c) and (d), the contact force curves and the corresponding confidence curves for constant desired forces are presented. The results demonstrate that higher desired forces yield greater and more stable confidence levels. However, when the desired force exceeds a certain threshold, the confidence level approaches saturation without further improvement. Moreover, excessive desired force may induce patient discomfort and cause excessive breast tissue deformation. As shown in Fig. 6(e), the variable desired force control strategy incorporates both CFS-DFP and UIC-DFU methods at different path points and between adjacent points respectively. During ultrasound scanning, this approach enables real-time online adjustment of desired force magnitude, achieving consistently high and stable confidence levels. In addition, the CNN-Swin Transformer hybrid model from [30] was successfully applied to perform intelligent lesion recognition and segmentation on the ultrasound images obtained by the variable desired force control method on the first phantom, as shown in Fig. 6(e-1) and (e-2). This demonstrates that the smoothed scanning path preserves sufficient image quality and completeness to enable clinically meaningful lesion segmentation. It indicates that the generated path is effective and that the amount of smoothing applied is not clinically detrimental. To more intuitively analyze the influence of different force control methods on the ultrasound image confidence, bar plots with error bars were generated, as shown in Fig. 6(f). As the desired force increased, the mean confidence values increased clearly, while the standard deviation of the confidence decreased noticeably. The mean and standard deviation of confidence values obtained using the proposed variable desired force control method under different paths were comparable to those obtained using a constant desired force of F_{ex}(8 N for the first phantom, 6 N for the second phantom, 3 N for the third phantom). The results indicate that the proposed method achieves probe-tissue contact quality comparable to constant force control (with higher desired forces), even while using lower contact forces.

V. Conclusion

This letter proposes the DO-ACDC, which effectively filters out local path anomalies. Compared to NFO and EBTSO, this approach generates smoother paths with reduced deviation and smaller angle increments, demonstrating superior optimization performance. On this basis, a variable desired force control method is further proposed, which clearly reduces desired force while maintaining intimate probe-tissue contact, thereby achieving a balance between patient comfort and probe-tissue contact quality.

In the future, we plan to address potential path planning errors caused by tissue deformation under probe compression by incorporating finite element methods for online path correction. Moreover, as our experimental validation was limited to three breast phantoms, we plan to conduct clinical ultrasound scanning experiments to evaluate the robustness and generalizability of the proposed method in real patient settings. Finally, we believe that the 6DOF manipulator is not the only ultimate form of automated ultrasound scanning, so it may be of great value to generalize the proposed algorithm to other robotic manipulators in the future.

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